

MINOT INTL, ND, USA

WMO: 727676

Lat: 48.255N

Lon: 101.273W

Elev: 1665

StdP: 13.83

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 24013

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-13.0	-24.9	1.5	-17.2	-20.1	1.9	-12.6	33.1	18.8	30.9	19.2	10.7	290	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	91.0	69.2	87.7	68.5	84.2	66.5	73.5	84.4	71.2	82.4	69.1	80.4	12.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	118.1	80.0	67.6	107.7	77.9	65.1	98.7	75.3	38.3	84.1	36.1	82.6	34.2	80.5	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.3	25.2	22.4	DB	-22.7	98.4	5.8	3.5	-26.9	101.0	-30.3	103.0	-33.5	105.0	-37.8	107.6	(4)
(5)			WB	-23.0	77.7	5.7	3.0	-27.1	79.8	-30.4	81.6	-33.6	83.3	-37.7	85.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	12.3	15.4	27.5	42.3	54.8	64.5	70.5	69.1	59.5	44.8	29.3	16.8		
	DBStd	23.39	15.12	14.09	14.01	10.94	8.75	6.77	5.76	6.48	8.44	9.67	12.14	13.86		
	HDD50	5055	1168	970	702	272	52	2	0	0	16	218	626	1029		
	HDD65	8733	1633	1390	1163	681	331	87	17	32	205	628	1072	1494		
	CDD50	2273	0	0	4	42	199	438	637	592	301	57	3	0		
	CDD65	477	0	0	0	1	14	73	189	159	39	2	0	0		
	CDH74	5282	0	0	0	28	239	788	1913	1741	532	41	0	0		
	CDH80	1872	0	0	0	5	61	225	688	705	179	9	0	0		
(14) Wind	WSAvg	11.0	12.2	11.7	11.6	11.6	11.6	10.3	9.4	9.4	10.1	11.4	11.6	11.7		
(15) Precipitation	PrecAvg	18.0	0.7	0.5	0.8	1.7	2.4	3.1	2.6	2.0	1.7	1.0	0.7	0.6		
	PrecMax	28.4	2.1	1.5	2.8	8.1	6.8	8.0	7.4	6.3	6.1	4.6	2.7	1.9		
	PrecMin	10.3	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.2	0.1	0.0	0.0	0.0		
	PrecStd	4.6	0.5	0.4	0.7	1.8	1.6	1.5	1.5	1.3	1.1	0.6	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	48.0	66.0	78.0	86.4	90.6	95.1	96.9	91.1	79.4	63.5	48.2		
		MCWB	38.5	40.7	49.4	55.8	62.6	71.0	72.0	70.2	66.0	57.9	48.6	39.4		
	2%	DB	41.1	42.5	57.3	70.9	80.3	85.9	90.4	91.5	84.8	71.4	55.8	43.1		
		MCWB	35.9	37.4	45.7	52.2	59.8	67.0	71.2	68.2	63.4	55.0	44.6	36.8		
	5%	DB	37.6	38.4	50.5	65.9	75.4	82.0	87.0	87.6	80.5	65.2	50.2	39.0		
		MCWB	33.9	34.7	42.2	49.5	56.7	65.2	70.4	67.7	61.8	51.3	41.3	34.1		
	10%	DB	33.8	34.6	45.2	60.7	71.5	78.5	83.7	83.6	75.1	59.9	45.3	35.6		
		MCWB	30.9	31.9	39.4	47.3	55.4	63.3	68.0	66.2	59.8	49.2	39.0	31.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.1	41.3	51.4	57.7	65.9	75.2	77.5	75.5	70.0	60.8	50.1	40.4		
		MCDB	44.6	46.7	62.8	74.9	79.3	84.6	87.8	86.3	84.3	73.3	62.9	46.5		
	2%	WB	36.3	37.9	46.6	53.9	62.3	70.5	74.4	71.7	66.6	56.3	45.5	37.2		
		MCDB	40.6	41.7	56.4	67.9	74.8	80.8	85.1	84.2	78.8	67.8	54.3	42.7		
	5%	WB	33.9	35.0	42.6	50.7	59.4	67.8	72.3	69.6	64.0	53.3	41.9	34.4		
		MCDB	37.6	38.2	50.0	63.0	70.8	78.0	82.9	82.3	75.7	62.8	49.1	38.8		
	10%	WB	31.0	32.2	39.4	48.1	57.1	65.2	70.2	67.9	61.3	50.2	39.0	31.7		
		MCDB	33.8	34.6	44.9	59.3	67.9	75.5	80.9	79.8	72.9	59.2	45.2	35.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	16.2	17.3	21.3	22.3	20.4	22.3	24.3	22.7	19.2	16.4	14.9		
		MCDBR	17.6	17.8	24.9	30.2	29.3	26.2	26.9	29.8	31.0	28.6	24.2	18.4		
	5% WB	MCWBR	14.3	13.8	15.8	15.7	13.9	12.3	12.2	12.2	14.0	15.1	14.9	13.9		
		MCWBR	17.2	16.5	24.0	27.8	24.4	22.9	24.0	26.4	27.0	26.6	23.2	18.6		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.258	0.292	0.329	0.346	0.367	0.376	0.385	0.341	0.299	0.266	0.244		
	taud		2.406	2.427	2.452	2.401	2.409	2.394	2.383	2.338	2.432	2.525	2.493	2.449		
	Ebn at Noon		268	292	297	293	291	284	279	271	274	272	260	257		
	Edh at Noon		20	25	29	34	36	36	36	36	30	23	19	17		
(44) All-Sky Solar Radiation	RadAvg		504	853	1233	1550	1826	1972	2096	1721	1273	797	481	355		
	RadStd		45	85	100	114	129	142	84	119	92	88	71	38		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	-1.10	-1.76	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page